

KILLAM AGDM, AB, Canada

WMO: 712280

Lat: 52.8475N

Lon: 111.8719W

Elev: 675

StdP: 93.48

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.9	-28.6	-35.6	0.1	-31.8	-32.2	0.2	-28.5	13.3	-6.6	11.6	-8.6	2.9	130	0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	29.1	17.1	27.2	16.5	25.4	15.7	19.3	26.1	18.0	24.6	17.0	23.1	4.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	13.0	22.6	15.6	12.0	21.0	14.5	11.1	19.7	57.9	26.0	53.6	24.8	50.1	23.1	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.6	8.5	DB	-36.3	32.4	3.4	2.3	-38.7	34.1	-40.8	35.4	-42.7	36.7	-45.2	38.4	(4)
(5)			WB	-36.5	22.0	3.3	1.3	-38.9	22.9	-40.9	23.6	-42.7	24.3	-45.2	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-11.7	-11.7	-5.0	3.6	10.5	14.6	16.9	15.5	10.7	3.8	-5.3	-11.6	(6)	
	DBStd	12.19	8.40	7.70	7.65	5.17	4.31	3.08	2.78	3.17	4.34	4.98	7.08	7.73	(7)	
	HDD10.0	3361	672	607	465	197	46	3	0	2	42	199	458	670	(8)	
	HDD18.3	5772	930	841	723	441	242	118	63	97	230	452	708	928	(9)	
	CDD10.0	663	0	0	0	6	63	141	213	172	63	6	0	0	(10)	
	CDD18.3	33	0	0	0	0	1	6	17	8	1	0	0	0	(11)	
	CDH23.3	912	0	0	0	3	74	122	328	266	116	3	0	0	(12)	
	CDH26.7	219	0	0	0	0	11	19	88	67	33	0	0	0	(13)	
(14) Wind	WSAvg	4.0	4.0	3.8	4.1	4.6	4.6	4.2	3.5	3.4	3.8	4.1	3.9	3.8	(14)	
(15) Precipitation	PrecAvg	408	21	13	18	30	42	73	71	55	33	18	18	17	(15)	
	PrecMax	547	61	24	43	71	104	110	151	125	83	48	61	30	(16)	
	PrecMin	261	4	2	1	4	10	17	18	6	1	4	3	2	(17)	
	PrecStd	85	13	7	10	16	24	26	30	24	22	10	13	7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	5.5	13.3	22.5	28.2	28.9	31.7	31.2	30.5	23.1	13.0	6.0	(19)	
		MCWB	2.1	2.3	6.9	10.3	13.5	16.6	19.0	16.9	15.2	12.0	6.5	2.7	(20)	
	2%	DB	3.6	3.3	8.7	19.4	25.4	26.6	29.1	28.4	26.6	18.2	8.4	3.5	(21)	
		MCWB	1.2	0.8	4.3	9.1	12.5	15.6	18.7	16.8	14.2	9.8	4.0	1.1	(22)	
	5%	DB	1.8	1.5	6.0	15.8	22.9	24.5	26.9	26.3	23.5	15.0	5.6	1.2	(23)	
		MCWB	0.0	-0.4	2.6	7.0	11.5	14.8	18.1	16.6	13.3	8.0	2.3	-0.8	(24)	
	10%	DB	-0.3	-0.4	3.9	12.9	20.3	22.3	24.8	24.2	20.0	12.1	3.4	-0.9	(25)	
		MCWB	-1.9	-1.9	1.4	5.6	10.2	13.8	17.1	16.1	11.8	6.6	1.0	-2.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	2.6	7.6	10.9	14.8	18.8	21.7	20.4	16.4	12.5	6.9	2.9	(27)	
		MCDB	4.6	5.3	12.5	21.8	25.2	26.0	27.8	25.5	26.9	21.6	11.6	5.5	(28)	
	2%	WB	1.3	1.0	4.6	9.1	13.2	17.1	20.0	18.7	14.8	10.4	4.5	1.2	(29)	
		MCDB	3.3	2.7	8.6	18.5	22.9	23.9	26.8	25.7	24.6	16.9	8.0	3.4	(30)	
	5%	WB	0.1	-0.3	2.7	7.5	12.2	15.7	18.7	17.4	13.6	8.7	2.6	-0.7	(31)	
		MCDB	1.8	1.5	5.6	14.7	21.2	22.0	25.3	24.2	22.2	14.2	5.1	1.1	(32)	
	10%	WB	-1.8	-2.0	1.5	6.2	11.0	14.7	17.6	16.3	12.4	7.0	1.2	-2.4	(33)	
		MCDB	-0.3	-0.4	3.7	12.3	19.0	20.8	23.6	23.0	19.3	11.6	3.3	-0.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	10.9	10.3	12.3	14.0	12.3	13.5	14.3	14.5	12.5	9.9	9.6	(35)	
		MCDBR	10.9	12.0	12.2	18.5	18.5	16.5	17.9	19.2	21.3	17.8	12.4	11.1	(36)	
	5% WB	MCWBR	9.2	9.9	8.4	9.5	8.0	7.3	9.0	9.2	10.1	10.2	8.5	9.2	(37)	
		MCWBR	11.1	11.6	11.7	16.4	16.6	13.9	16.1	16.5	18.9	17.0	11.6	11.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.264	0.285	0.298	0.332	0.352	0.367	0.362	0.375	0.317	0.282	0.256	0.242	(40)	
		taud	2.284	2.285	2.382	2.358	2.352	2.344	2.387	2.328	2.456	2.533	2.449	2.329	(41)	
	Ebn at Noon		731	827	898	902	898	884	882	846	867	839	758	703	(42)	
		Edh at Noon	60	83	93	109	115	118	111	111	86	64	51	48	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.05	2.11	3.58	4.72	5.78	6.00	6.27	5.12	3.58	2.12	1.06	0.78	(44)	
	RadStd		0.10	0.16	0.27	0.44	0.39	0.41	0.32	0.39	0.38	0.27	0.11	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (82 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page